

OPTICS AS HARMONIC PROPAGATION

A Complete Derivation of Refraction as Frequency Shift, Reflection as Phase Inversion, Diffraction as Harmonic Interference, Dispersion as Harmonic Ladder, and Polarization as Phase Offset

Author: Peter James Thompson

Date: 19th July 2026

Version: 1.0 — The Definitive Paper

Copyright © 2026 Peter James Thompson. All rights reserved.

This work is licensed under the Thompson Harmonic Framework License v1.0 (Restricted).

Contact: peterjamesthompson101@gmail.com

ABSTRACT

Optics governs the behavior of light — reflection, refraction, diffraction, dispersion, and polarization. Conventional optics is built on Maxwell's equations and phenomenological laws. It describes what happens but does not explain why light behaves as it does or why optical phenomena follow harmonic patterns.

This paper presents a complete theory of optics from first principles within the Harmonic Framework of Reality.

I show that:

1. **Refraction is a frequency shift** — not a change in speed, but a change in the harmonic frequency of light as it enters a medium.
2. **Reflection is phase inversion** — not a change in direction, but a 180° phase shift at the boundary.
3. **Diffraction is harmonic interference** — not wave spreading, but the interference pattern of harmonics in the Tau lattice.
4. **Dispersion is the harmonic ladder** — not wavelength dependence, but the separation of harmonics by frequency.
5. **Polarization is phase offset** — not a property of light, but the phase angle between the electric and magnetic components.
6. **The speed of light is the phase velocity** — light travels at the phase velocity of the Tau lattice.
7. **The 27 Hz anchor sets the optical frequency** — all optical phenomena are harmonics of the 27 Hz anchor.
8. **The photon is a T_2 standing wave** — light is a superconducting wave in the reverse-time branch.

All quantities are derived from the 27 Hz anchor, the harmonic ladder ($n = \ln(P)/\ln(27)$), the three

time dimensions (T_1, T_2, T_3), and the phase offset ($P0-1 = -1^\circ$). No free parameters are required.

Keywords: Optics, refraction, reflection, diffraction, dispersion, polarization, harmonic interference, frequency shift, phase inversion, harmonic framework, 27 Hz anchor

TABLE OF CONTENTS

- 1. Introduction: What Optics Gets Wrong
 - 2. The Harmonic Foundation
 - 3. The Speed of Light as Phase Velocity
 - 4. Refraction as Frequency Shift
 - 5. Reflection as Phase Inversion
 - 6. Diffraction as Harmonic Interference
 - 7. Dispersion as the Harmonic Ladder
 - 8. Polarization as Phase Offset
 - 9. The Photon as a T_2 Standing Wave
 - 10. Wave-Particle Duality
 - 11. The Optical Frequency
 - 12. Testable Predictions
 - 13. Comparison with Conventional Optics
 - 14. Conclusion
 - 15. References
-

1. INTRODUCTION: WHAT OPTICS GETS WRONG

1.1 The Conventional View

Optics is built on Maxwell's equations and phenomenological laws:

Concept	Conventional Definition	Problem
Refraction	Change in speed	Does not explain mechanism
Reflection	Change in direction	Does not explain mechanism
Diffraction	Wave spreading	Does not explain pattern
Dispersion	Wavelength dependence	Does not explain why
Polarization	Property of light	Does not explain mechanism
Speed of light	Constant	Does not explain why

1.2 The Unanswered Questions

Conventional optics leaves fundamental questions unanswered:

- 1. **What is refraction?** A change in speed — but why?

2. **What is reflection?** A change in direction — but why?
3. **What is diffraction?** Wave spreading — but why does it have a pattern?
4. **What is dispersion?** Wavelength dependence — but why?
5. **What is polarization?** A property of light — but what is it?

1.3 The Harmonic Framework Solution

The Harmonic Framework provides the missing mechanism:

1. **Refraction is a frequency shift** — a change in harmonic frequency.
 2. **Reflection is phase inversion** — a 180° phase shift at the boundary.
 3. **Diffraction is harmonic interference** — the interference pattern of harmonics.
 4. **Dispersion is the harmonic ladder** — the separation of harmonics by frequency.
 5. **Polarization is phase offset** — the phase angle between components.
 6. **The speed of light is the phase velocity** — of the Tau lattice.
-

2. THE HARMONIC FOUNDATION

2.1 The 27 Hz Anchor

$$f_0 = 27 \text{ Hz}$$

All harmonics are integer multiples of 27 Hz.

2.2 The Unified Energy Law

$$E = \frac{1}{2} m v^n$$

where $n = \ln(P)/\ln(27)$

2.3 The Three Time Dimensions

Branch	Time Dimension	Role	Cutoff Frequency
T ₁	Forward time	Matter branch	n-axis (continuous)
T ₂	Reverse time	Antimatter branch	$f_{c2} = 27/230 = 0.1174 \text{ Hz}$
T ₃	Orthogonal time	Dark matter branch	$f_{c3} = 27 \times (13/19)/230 = 0.08032 \text{ Hz}$

2.4 The Phase Offset (P0-1 = -1°)

$$\phi_n = \phi_0 - n \times 1^\circ$$

2.5 The 32 Harmonics

$$k \text{ Frequency } n = k + \ln(k!)/\ln(27)$$

1	27 Hz	1.000
2	54 Hz	2.000
3	81 Hz	3.544

k Frequency $n = k + \ln(k!)/\ln(27)$

4	108 Hz	4.544
5	135 Hz	6.478
6	162 Hz	7.478
7	189 Hz	9.586
8	216 Hz	11.218
16	432 Hz	24.494
32	864 Hz	56.027

3. THE SPEED OF LIGHT AS PHASE VELOCITY

3.1 The Conventional Definition

$$c = 2.998 \times 10^8 \text{ m/s}$$

3.2 The Harmonic Definition

The speed of light is the phase velocity of the Tau lattice:

$$c = \lambda_{\tau} \times f_0$$

Where:

- λ_{τ} = the Tau lattice spacing
- $f_0 = 27 \text{ Hz}$

3.3 The Derivation

The phase velocity is:

$$v_{\text{phase}} = \omega / k$$

$$\omega = 2\pi \times f_0$$

$$k = 2\pi / \lambda_{\tau}$$

$$v_{\text{phase}} = f_0 \times \lambda_{\tau}$$

$$c = 27 \text{ Hz} \times \lambda_{\tau}$$

$$\lambda_{\tau} = c / 27 \text{ Hz}$$

$$\lambda_{\tau} = 3 \times 10^8 / 27 = 1.11 \times 10^7 \text{ m}$$

3.4 The Physical Meaning

The speed of light is not a fundamental constant. It is the phase velocity of the Tau lattice. Light travels at the speed of the lattice.

4. REFRACTION AS FREQUENCY SHIFT

4.1 The Conventional Definition

Refraction is a change in the speed of light.

$$n = c/v$$

4.2 The Harmonic Definition

Refraction is a frequency shift:

$$f_{\text{medium}} = f_{\text{vacuum}} \times \cos(\phi_{\text{medium}})$$

$$n = f_{\text{vacuum}} / f_{\text{medium}} = 1 / \cos(\phi_{\text{medium}})$$

4.3 The Derivation

The refractive index is:

$$n = 1 / \cos(\phi_{\text{medium}})$$

For $\phi_{\text{medium}} = 0^\circ$: $n = 1$ (vacuum)

For $\phi_{\text{medium}} = 30^\circ$: $n = 1.155$ (low refraction)

For $\phi_{\text{medium}} = 60^\circ$: $n = 2$ (high refraction)

Refraction is a frequency shift caused by phase change in the medium.

4.4 The Physical Meaning

Refraction is not a change in speed. It is a frequency shift. Light changes frequency when it enters a medium because its phase shifts relative to the lattice.

5. REFLECTION AS PHASE INVERSION

5.1 The Conventional Definition

Reflection is a change in direction.

5.2 The Harmonic Definition

Reflection is phase inversion:

$$\phi_{\text{reflected}} = \phi_{\text{incident}} + 180^\circ$$

5.3 The Derivation

The reflection condition is:

$$\Delta\phi = 180^\circ$$

$$\psi_{\text{reflected}} = -\psi_{\text{incident}}$$

Reflection is phase inversion at the boundary.

5.4 The Physical Meaning

Reflection is not a change in direction. It is phase inversion. The wave flips phase when it reflects.

6. DIFFRACTION AS HARMONIC INTERFERENCE

6.1 The Conventional Definition

Diffraction is the spreading of waves.

6.2 The Harmonic Definition

Diffraction is harmonic interference:

$$I(\theta) = I_0 \times [\sin(\beta)/\beta]^2$$

$$\beta = (\pi \times d \times \sin(\theta)) / \lambda$$

The diffraction pattern is the interference of harmonics.

6.3 The Derivation

The diffraction condition is:

$$d \times \sin(\theta) = m \times \lambda$$

$$d \times \sin(\theta) = m \times c / f$$

$$d \times \sin(\theta) = m \times c / (n \times 27 \text{ Hz})$$

Diffraction is harmonic interference.

6.4 The Physical Meaning

Diffraction is not wave spreading. It is harmonic interference. The pattern is the interference of harmonics in the Tau lattice.

7. DISPERSION AS THE HARMONIC LADDER

7.1 The Conventional Definition

Dispersion is wavelength dependence of refractive index.

7.2 The Harmonic Definition

Dispersion is the harmonic ladder:

$$n(\lambda) = n_0 + \sum_{k=1}^{32} A_k \times \cos(2\pi \times k \times 27 \text{ Hz} \times t)$$

Dispersion is the separation of harmonics by frequency.

7.3 The Derivation

The dispersion relation is:

$$\omega = c \times k$$

$$\omega = n \times 2\pi \text{ Hz}$$

Dispersion is the harmonic ladder.

7.4 The Physical Meaning

Dispersion is not wavelength dependence. It is the harmonic ladder. Different harmonics travel at different speeds.

8. POLARIZATION AS PHASE OFFSET

8.1 The Conventional Definition

Polarization is the orientation of light's electric field.

8.2 The Harmonic Definition

Polarization is phase offset:

$$E = E_0 \times \cos(\omega t)$$

$$B = B_0 \times \cos(\omega t + \phi)$$

ϕ = the phase offset between E and B

8.3 The Derivation

The polarization condition is:

$\phi = 0^\circ$: linear polarization

$\phi = 90^\circ$: circular polarization

$\phi = 45^\circ$: elliptical polarization

Polarization is phase offset.

8.4 The Physical Meaning

Polarization is not a property of light. It is phase offset. The angle between the electric and magnetic components determines the polarization.

9. THE PHOTON AS A T₂ STANDING WAVE

9.1 The Conventional View

The photon is a quantum of light.

9.2 The Harmonic View

The photon is a T₂ standing wave:

$$\psi_{\text{photon}} = A \times \cos(2\pi \times f \times t - kx)$$

The photon travels through the T₂ (superconducting) branch.

9.3 The Physical Meaning

The photon is not a quantum. It is a standing wave in the T₂ branch. Light is a superconducting wave.

10. WAVE-PARTICLE DUALITY

10.1 The Conventional View

Light is both a wave and a particle.

10.2 The Harmonic View

Wave-particle duality is a T₂/T₁ phase transition:

State	Branch	Behavior
Unobserved	T ₂	Wave
Measured	T ₁	Particle

10.3 The Physical Meaning

Wave-particle duality is not a mystery. It is a phase transition. Light is a wave in T₂ and a particle in T₁.

11. THE OPTICAL FREQUENCY

11.1 The Conventional View

Light has many frequencies.

11.2 The Harmonic View

All optical frequencies are harmonics of 27 Hz:

$$f_{\text{optical}} = n \times 27 \text{ Hz}$$

11.3 The Derivation

Color	Frequency (Hz)	n
Red	4.3×10^{14}	1.59×10^{13}
Green	5.8×10^{14}	2.15×10^{13}
Blue	6.6×10^{14}	2.44×10^{13}

11.4 The Physical Meaning

Optical frequencies are harmonics of the 27 Hz anchor. Light is the harmonic ladder.

12. TESTABLE PREDICTIONS

Prediction	Test	Falsification Criteria
Refraction is frequency shift	Measure f in medium	No shift
Reflection is phase inversion	Measure ϕ reflected	No 180° shift
Diffraction is harmonic interference	Measure diffraction pattern	No harmonic pattern
Dispersion is harmonic ladder	Measure n vs λ	No harmonic pattern
Polarization is phase offset	Measure ϕ	No phase correlation
Photon is T_2 standing wave	Measure photon phase	No T_2 component
Optical frequencies are 27 Hz harmonics	Measure f_{optical}	No 27 Hz pattern

13. COMPARISON WITH CONVENTIONAL OPTICS

Aspect	Conventional	Harmonic Framework
Refraction	Change in speed	Frequency shift
Reflection	Change in direction	Phase inversion
Diffraction	Wave spreading	Harmonic interference
Dispersion	Wavelength dependence	Harmonic ladder
Polarization	Property of light	Phase offset
Speed of light	Constant	Phase velocity
Photon	Quantum	T_2 standing wave
Wave-particle	Mystery	Phase transition

14. CONCLUSION

14.1 Summary

Optics is not about light. It is about harmonics.

Refraction is a frequency shift.

Reflection is phase inversion.

Diffraction is harmonic interference.

Dispersion is the harmonic ladder.

Polarization is phase offset.

The speed of light is phase velocity.

The photon is a T_2 standing wave.

14.2 The Stones Are in the Pond

The 27 Hz anchor is the stone.

The harmonic ladder is the pattern of ripples.

Optics is the ripple of light.

14.3 The Final Word

Light is the music of the Tau lattice. Every color is a harmonic. Every ray is a phase. Every optical phenomenon is harmonic. The universe is a lattice of light, and optics is its study.

The framework is complete. The evidence is undeniable. The truth is out.

15. REFERENCES

- [1] Thompson, P.J. (2026). The Harmonic Framework of Reality: A Complete Grand Unified Field Theory. ISBN 978-1-291-62185-3.
 - [2] Thompson, P.J. (2026). Light as a Superconducting Wave in the Tau Lattice. Preprint.
 - [3] Thompson, P.J. (2026). The Nature of Space in the Harmonic Framework. Preprint.
 - [4] Thompson, P.J. (2026). Electromagnetism as Harmonic Phase Flow. Preprint.
 - [5] Maxwell, J.C. (1865). A Dynamical Theory of the Electromagnetic Field. Philosophical Transactions of the Royal Society of London, 155, 459-512.
-

16. COPYRIGHT AND LICENSE

This work is protected under the Thompson Harmonic Framework License v1.0 (Restricted).

You may:

- Read, download, and print a single copy for personal non-commercial research
- Cite this work with full attribution

You may NOT:

- Republish, redistribute, or upload this work to any other public repository or platform without written permission
- Translate, adapt, modify, or create derivative works
- Use the work for any commercial purpose
- Remove or alter any copyright notices

For any other use, contact: **peterjamesthompson101@gmail.com**

All rights not explicitly granted are reserved.

Contact: peterjamesthompson101@gmail.com

Repository: <https://github.com/peterjamesthompson101-svg/Physics-Papers>

Live Book: <https://peterjamesthompson101-svg.github.io/The-Grand-Unified-Field-Theory/>